University of Science and Technology of Hanoi

Fundamentals of Data Science

USA Flight Delay Analysis

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Hanoi, Sep 2025

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ACKNOWLEDGEMENTS

We would like to express our sincere gratitude to our supervisor, Nguyen Duc Dung, for their valuable guidance, feedback, and continuous encouragement throughout the development of this project. Their insights greatly helped us refine our approach and achieve meaningful results.

We also appreciate the support from our peers and colleagues who provided constructive discussions and suggestions during the research process. Their perspectives enriched our understanding of the subject matter.

ABSTRACT

Flight delays pose a significant operational and economic challenge for the aviation industry, affecting both airline efficiency and passenger experience. This project examines the factors influencing flight departure delays by integrating historical flight records with meteorological data for major airports in Washington State. A data engineering pipeline was developed to collect, clean, and merge data from the Bureau of Transportation Statistics (BTS) and the National Oceanic and Atmospheric Administration (NOAA). After comprehensive preprocessing and feature engineering, several machine learning models were employed to analyze delay patterns and identify key predictors. Statistical analysis and visualization techniques were applied to explore relationships between time, weather, and route characteristics under different delay scenarios. The findings provide insights into the underlying dynamics of flight delays and demonstrate how data-driven approaches can enhance predictive capability and operational decision-making in the aviation sector.

Keywords: *Flight Delay Prediction, Data Engineering, Machine Learning, Weather Integration, Statistical Analysis, Feature Engineering, Aviation Analytics.*

Chapter 1

Introduction

1.1 Business Analysis

1.1.1 Background

Flight delays are a persistent challenge in the aviation industry, leading to significant operational, financial, and customer-service impacts. Even minor delays can cascade across the network, disrupting schedules, increasing costs, and reducing passenger satisfaction. With the expansion of air traffic and the growing influence of external factors such as weather and airport congestion, understanding and predicting flight delays has become an essential task for both airlines and transportation authorities. The integration of data science and machine-learning techniques provides new opportunities to analyze historical data and uncover complex relationships between flight, weather, and operational variables.

1.1.2 Problem Statement

Flight operations are influenced by multiple interacting factors—weather conditions, scheduling patterns, route distances, and airline operations. Traditional statistical approaches often fail to capture the non-linear dependencies and dynamic patterns that drive delay events. This study aims to investigate the key factors contributing to flight departure delays and to develop a data-driven analytical framework that can detect patterns and relationships across diverse datasets. By merging flight performance data with meteorological information, the study seeks to identify major predictors of delay and examine how environmental and temporal factors affect flight punctuality.

1.1.3 Analytical Approach

This project employs a data engineering and machine-learning pipeline to process and analyze flight and weather data. Datasets from the Bureau of Transportation Statistics (BTS) and the National Oceanic and Atmospheric Administration (NOAA) were collected, cleaned, and merged through automated data-processing steps. Feature engineering was applied to transform time-based and weather-related variables into model-ready formats. Descriptive statistics, correlation analysis, and visualization techniques were used to explore relationships among variables, while classification algorithms such as Logistic Regression, Random Forest, and LightGBM were implemented to model delay likelihood. Insights derived from these analyses contribute to understanding the determinants of flight delay behavior.

1.1.4 Goals

The goal of this study is to build a comprehensive analytical system capable of predicting flight departure delays and identifying their main causes. Specifically, it aims to integrate flight and weather data into a unified dataset, construct machine-learning models to analyze delay patterns, and generate interpretable insights that can support airline operations and scheduling decisions. Through this approach, the project bridges data engineering and data science to enhance the reliability and efficiency of air-transport services.

1.1.5 Implications

The outcomes of this analysis can assist airlines, airports, and regulators in improving operational planning and resource allocation. By recognizing weather-related and temporal factors that most strongly affect delay risk, stakeholders can design early-warning systems, optimize flight scheduling, and enhance passenger communication. Moreover, the integration of predictive modeling into decision-support systems demonstrates how data-driven methods can contribute to more resilient and efficient aviation management.

Chapter 2

Data Collection and Preparation

2.1 Introduction

This project employs two distinct datasets — one containing flight operation records and another containing hourly weather observations — to support the analysis and prediction of flight departure delays. The use of both datasets is essential because flight performance is not only influenced by operational factors such as scheduling and routes, but also by environmental conditions like temperature, wind, and visibility. By combining these two perspectives, the study aims to capture a more realistic representation of the factors contributing to flight delays.

Therefore, each dataset was processed and cleaned individually before being merged into a single analytical dataset that serves as the foundation for further exploration and modeling.

2.1.1 Flight Dataset

The flight dataset used in this project is the **Airline On-Time Performance Data**, obtained from the **U.S. Bureau of Transportation Statistics**, which provides detailed records of domestic flight operations across the United States. It contains information about each scheduled flight, including the airline carrier, origin and destination airports, scheduled and actual departure times,... This dataset serves as the primary source for analyzing operational factors contributing to departure delays.

For this study, only flights departing from Washington State during the second quarter (Q2) were used. The dataset consists of 65,374 entries and 17 features, and is provided in CSV (Comma-Separated Values) format.

MONTH	DAY_OF_MONTH	OP_UNIQUE_CARRIER	ORIGIN	CRS_ARR_TIME	DEST	...	DISTANCE	DEP_DELAY	DEP_DEL15
4	1	AA	SEA	9.33	SEA	...	2279.0	11.0	0
4	1	UA	GEG	18.12	LAS	...	224.0	18.0	1
4	1	AS	SEA	15.32	BUR	...	937.0	20.0	1
...	...	...	...	...	...	...	...	...	...
6	30	WN	SEA	22.75	SMF	...	605.0	-5.0	0
6	30	WN	SEA	16.92	STL	...	1709.0	10.0	0

Table 2.1: Airline On-Time Performance Dataset

The data types and detailed discussion of each feature in the dataset are provided below:

Features	**Type**	**Data Type**	**Explanation**
MONTH	Temporal	Integer	Month of the flight date.
DAY_OF_MONTH	Temporal	Integer	Day of the month.
DAY_OF_WEEK	Temporal	Integer	Day of the week.
OP_UNIQUE_CARRIER	Categorical	String	Unique carrier code identifying the airline.
ORIGIN	Categorical	String	Code of the departure (origin) airport.
ORIGIN_STATE_ABR	Categorical	String	Two-letter abbreviation of the origin state.
DEST	Categorical	String	Code of the arrival (destination) airport.
DEST_STATE_ABR	Categorical	String	Two-letter abbreviation of the destination state.
CRS_DEP_TIME	Temporal	Integer	Scheduled departure time (in HHMM format).
CRS_ARR_TIME	Temporal	Integer	Scheduled arrival time (in HHMM format).
CANCELLED	Quantitative (Binary)	Float	Indicates if the flight was cancelled (1 = Yes, 0 = No).
CRS_ELAPSED_TIME	Quantitative	Float	Scheduled flight time in minutes.
DISTANCE	Quantitative	Float	Distance between origin and destination airports (in miles).
DISTANCE_GROUP	Categorical (Ordinal)	Integer	Group number representing distance range.
DEP_DELAY	Quantitative	Float	Difference between actual and scheduled departure time.
DEP_DEL15	Quantitative (Binary)	Integer	1 if departure delay ¿ 15 minutes, otherwise 0.
DEP_DELAY_GROUP	Categorical (Ordinal)	Float	Grouping of departure delay intervals.

Table 2.2: Features Explanation

2.1.2 Weather Dataset

The weather dataset used in this project was obtained from the **National Oceanic and Atmospheric Administration (NOAA)**, which provides historical hourly weather observations recorded at airports across the United States. To ensure consistency with the flight dataset, only data from major airports in Washington State during the second quarter (Q2) were collected. The dataset contains over 50,000 hourly records with many key meteorological, stored in CSV format, with timestamps adjusted to local time zones to enable accurate integration with flight schedules.

STATION	**DATE**	**HourlyDewPointTemperature**	**HourlyDryBulbTemperature**	**HourlyRelativeHumidity**	**HourlyVisibility**	**HourlyWindSpeed**
72784524163	2025-04-01T00:53:00	35	41	79	10	10
72784524163	2025-04-01T01:53:00	34	41	76	10	0
...	...	...	...	...	...	...
72793724222	2025-06-30T23:53:00	51	64	63	10	3
72793724222	2025-06-30T18:53:00	52	69	55	10	9

Table 2.3: NOAA Dataset

The data types and detailed discussion of each feature in the dataset are provided below:

STATION	Categorical	String	Identifier code where the weather data was recorded.
DATE	Temporal	Datetime	Timestamp representing the date and hour of the observation.
HourlyDewPointTemperature	Quantitative	Float	Dew point temperature (°F).
HourlyDryBulbTemperature	Quantitative	Float	Air temperature (°F).
HourlyRelativeHumidity	Quantitative	Float	Percentage of relative humidity in the air (%).
HourlyVisibility	Quantitative	Float	Horizontal visibility distance (miles).
HourlyWindSpeed	Quantitative	Float	Average wind speed during the hour (miles per hour).

Table 2.4: Features Explanation

2.2 Data Preparation and Cleaning

2.2.1 Flight Data Cleaning and Preprocessing

To ensure the data is suitable for analysis and modeling, several preprocessing steps were performed:
Filtering by origin state:

- Kept only flights with ORIGIN_STATE_ABR = 'WA', since the state has five main airports, which simplifies the merging process with weather data.
- After this step, a total of 50372 flights departing from WA during Q2 were recorded.

Handling Missing and Duplicate Records:

- Removed all rows with missing or zero values, except for time-related variables (DEP_DELAY, DEP_DEL15, DEP_DELAY_GROUP, DEP_DELAY_NEW).
- Duplicate records were identified and removed to maintain data integrity.

Remove cancelled flight:

- Filtered out all cancelled flights (CANCELLED = 1) to focus on actual departure delays.
- Remove CANCELLED column.

Feature Engineering:

- CRS_DEP_TIME and CRS_ARR_TIME were converted from HHMM to decimal-hour format.
- New feature ORIGIN-DEST was created to represent flight routes for better analytical insight.

General:

- After prepareing, dataset have 17 features and 50186 samples.
- Dataset is ready for merging and analysis.

2.2.2 Weather Data Cleaning and Preprocessing

The weather dataset was cleaned and standardized to ensure accuracy and consistency before merging with the flight data **Handling Missing and Duplicate Records:**

- Removed records containing NULL values.

- Duplicate records were identified and removed.

Extract new features:

- Converted the DATE column to datetime format to enable time-based operations and ensure consistent timestamp handling.
- Extracted the day and month information into a new column date, formatted as “MM-DD” for easier temporal alignment.
- Extracted the hour component into a new column hour, which was later used to merge weather data with flight departure times.

Map airport codes:

- The STATION codes were mapped from NOAA station IDs to corresponding airport IATA codes to ensure consistency with the flight dataset during merging.

General:

- After prepareing, dataset have 8 features and 10851 samples.
- Dataset is ready for merging and analysis.

2.2.3 Merging Datasets

After cleaning and preprocessing both datasets, they were merged to create a comprehensive dataset for analysis:

Prepare Time Columns:

- Extracted the departure hour (DEP_HOUR) from CRS_DEP_TIME by flooring the scheduled time to the nearest hour.
- Created a new column FL_DATE by combining MONTH and DAY_OF_MONTH, with the year temporarily set to 2025.
- In the weather dataset, added the year “2025” to the date column (if missing) and converted it to datetime format.
- These transformations standardized date and time formats across both datasets, enabling accurate merging by hour and location.

Merge Flight & Weather Data

- Merged flight and weather datasets using a left join on matching keys: ORIGIN, FL_DATE, DEP_HOUR -¿ STATION, date, hour.
- Ensured each flight matched its corresponding hourly weather record.
- Removed intermediate columns that were no longer needed: [”DEP_HOUR”, ”FL_DATE”, ”STATION”, ”date”, ”hour”].
- The merged dataset contained 50,029 rows and 22 columns.

2.3 Feature Engineering

2.4 Final Dataset Overview

MONTH	DAY_OF_MONTH	ORIGIN	HourlyVisibility	HourlyWindSpeed	...	DEST	DISTANCE	DEP_DELAY	DEP_DEL15
4	1	GEG	5.0	10.0	...	PHX	1020.0	-5.0	0
4	1	GEG	10.0	10.0	...	PHX	1660.0	15.0	1
...	...	...	...	...	...	...	...	...	...
6	30	SEA	10.0	14.0	...	SMF	605.0	-5.0	0
6	30	SEA	10.0	9.0	...	STL	1709.0	10.0	0

Table 2.5: Final Dataset

Chapter 3

Statistics

3.1 Descriptive Statistics

Descriptive statistics summarize the central tendency and dispersion of the six core features used in this study: Monthly Return, Real Return, P/E Ratio, PE10, 12-month Volatility, and Dividend Yield. We compute descriptive statistics for the six core features both at the full-sample level and separately for major crisis periods. This step helps to identify the general characteristics of the S&P 500 index and related financial metrics before moving into deeper analysis.

3.1.1 Full Sample

Feature	Mean	Median	Std	Min	25%	50%	75%	Max
Monthly Return	0.011	0.008	0.084	-0.360	-0.022	0.008	0.034	0.399
Real Return	0.008	0.007	0.081	-0.365	-0.023	0.007	0.033	0.431
P/E Ratio	21.617	19.641	13.055	6.788	15.671	19.641	23.807	123.731
PE10	25.825	25.885	8.873	8.400	20.560	25.885	30.902	48.110
Volatility (12M)	0.274	0.264	0.123	0.054	0.182	0.264	0.335	0.686
Dividend Yield	0.025	0.021	0.011	0.011	0.018	0.021	0.032	0.062

Table 3.1: Descriptive statistics of the six main features (full sample).

- As shown in Table 3.1, returns average less than 1% per month but display high variability, with extremes near –36% and +40%. This reflects the high sensitivity of the S&P 500 to crisis events.
- Valuation ratios are elevated on average, with the P/E occasionally spiking above 120, while PE10 remains consistently high around 26.
- Volatility averages 27% annually but rises sharply in crises. By contrast, the dividend yield is relatively stable (mean of 2.5%), suggesting that its contribution to total returns is modest but steady across time.

Overall, these descriptive statistics highlight the combination of high return variability, elevated valuations, and episodic volatility spikes that characterize the U.S. equity market over the sample period.

3.1.2 Period Analysis

To capture structural changes in the U.S. equity market, the sample is divided into four periods: 1980–1989, 1990–1999, 2000–2009, and 2010–2023. Each decade reflects distinct market conditions shaped by major crises: the 1987 crash in the 1980s, the late-1990s dot-com bubble, the dot-com bust (2000–2002) and the Global Financial Crisis (2007–2009) in the 2000s, and the COVID-19 shock (2020) in the 2010s–2020s. This division allows comparison of returns, valuation, volatility, and dividends across both growth phases and crisis episodes.

- **1980–1989**

Feature	**Mean**	**Std**	**Min**	**Max**
Monthly Return	0.0129	0.0799	-0.1782	0.2564
Real Return	0.0080	0.0690	-0.1708	0.2054
P/E Ratio	12.0342	3.4787	6.7882	21.4174
PE10	14.3445	3.9348	8.4000	22.8500
Volatility (12M)	0.2673	0.0584	0.1078	0.4135
Dividend Yield	0.0420	0.0089	0.0261	0.0624

Table 3.2: Descriptive statistics for key features, 1980–1989.

  - Monthly and real returns averaged about 1.3% and 0.8% per month, respectively, equivalent to strong annual gains. However, volatility was high, with losses up to –18% and gains above 25%.
  - Valuation remained low compared to later decades (P/E $\approx$ 12, PE10 $\approx$ 14), reflecting relatively cheap equity prices in the 1980s.
  - Volatility averaged 26–27% annually, peaking above 40% during the 1987 market crash.
  - Dividend yield was the highest of all periods (mean $\approx$ 4.2%), indicating that dividends played a significant role in total returns.

  Overall, the 1980s combined strong growth and attractive valuations with occasional extreme shocks, most notably the Black Monday crash of 1987.

- **1990-1999**

  - Returns were the strongest among all periods (monthly $\approx$ 1.5% nominal, 1.2% real), though volatility remained high with swings from –19.6% to +28%.
  - Valuations expanded sharply (P/E $\approx$ 22, PE10 $\approx$ 29), reflecting the build-up of the dot-com bubble by the late 1990s.

Feature	Mean	Std	Min	Max
Monthly Return	0.0148	0.0790	-0.1961	0.2806
Real Return	0.0120	0.0734	-0.1806	0.2560
P/E Ratio	21.6416	5.1534	14.2141	33.9980
PE10	28.9100	8.4463	17.5900	48.1100
Volatility (12M)	0.2685	0.1120	0.0639	0.4903
Dividend Yield	0.0244	0.0073	0.0117	0.0388

Table 3.3: Descriptive statistics for key features, 1990–1999.

- Volatility averaged 27% and peaked near 49%, showing episodes of market stress despite the bull run.
- Dividend yield declined to around 2.4%, much lower than the 1980s, as equity prices outpaced dividend growth.

Overall, the 1990s combined exceptional equity performance with rising valuations and falling dividends, culminating in the late-decade dot-com bubble.

- **2000-2009**

Feature	Mean	Std	Min	Max
Monthly Return	0.0037	0.1072	-0.3595	0.3985
Real Return	0.0019	0.1098	-0.3648	0.4307
P/E Ratio	31.3906	22.0868	16.6149	123.7308
PE10	28.7900	7.0770	14.5900	47.5900
Volatility (12M)	0.3178	0.1947	0.0545	0.6856
Dividend Yield	0.0179	0.0050	0.0111	0.0360

Table 3.4: Descriptive statistics for key features, 2000–2009.

- Returns were weak (monthly $\approx$ 0.4% nominal, 0.2% real) with large variability, including extreme losses near –36% and gains above +40%.
- Valuations were elevated and unstable (P/E $\approx$ 31, PE10 $\approx$ 29), with the P/E ratio spiking above 120 during earnings collapses.
- Volatility averaged 32% annually and reached nearly 69%, the highest among all periods, reflecting the dot-com bust and the Global Financial Crisis.
- Dividend yield declined further to about 1.8%, offering limited income support to investors.

Overall, the 2000s were marked by two major crises, producing poor equity returns, extreme volatility, and unstable valuations, often referred to as a "lost decade" for U.S. equities.

- **2010-2023**

Feature	Mean	Std	Min	Max
Monthly Return	0.0109	0.0712	-0.2029	0.2303
Real Return	0.0086	0.0686	-0.2543	0.2540
P/E Ratio	21.4586	4.7640	13.4960	39.2575
PE10	29.8483	5.0396	21.7100	42.0500
Volatility (12M)	0.2520	0.0825	0.0845	0.4512
Dividend Yield	0.0189	0.0022	0.0129	0.0230

Table 3.5: Descriptive statistics for key features, 2010–2023.

  - Returns recovered (monthly $\approx$ 1.1% nominal, 0.9% real), close to long-term averages.
  - Valuations stayed elevated (P/E $\approx$ 21, PE10 $\approx$ 30), reflecting sustained optimism.
  - Volatility averaged 25% and spiked above 45%, notably during the COVID-19 shock in 2020.
  - Dividend yield remained low (mean $\approx$ 1.9%), continuing the declining trend from earlier decades.

Overall, the 2010s–2020s featured strong post-crisis recovery and high valuations, but with renewed turbulence during global shocks such as COVID-19.

3.2 Distribution Analysis

This section explores the distribution of the six core features using histograms that highlights skewness, fat tails, and outliers.

Figure 3.1: Distributions of the six core features: Monthly Return, Real Return, P/E Ratio, PE10, Volatility (12M), and Dividend Yield.

- Monthly and real returns are centered near zero but display fat tails and frequent outliers, indicating the presence of extreme market swings.
- Valuation ratios are right-skewed, with P/E showing extreme spikes when earnings collapsed, while PE10 is more stable but still biased toward high values.
- Volatility (12M) is positively skewed, with crisis-driven spikes reaching unusually high levels.
- Dividend yield shows the narrowest distribution, mostly stable around 2–3%, with occasional jumps above 5% during market downturns.

Overall, distributions confirm non-normal behavior in returns, persistent high valuations, crisis-induced volatility, and relatively stable income yields.

3.3 Correlation

To examine linear relationships among the six core features, we compute the Pearson correlation matrix and visualize it using a heatmap. This analysis helps identify potential dependencies, overlaps, and contrasts between returns, valuation metrics, volatility, and dividend yield.

Figure 3.2: Correlation matrix of the six main features.

- Monthly and real returns are almost perfectly correlated ($\rho \approx 0.97$) since real returns are inflation-adjusted versions of nominal returns.
- Valuation metrics and dividend yield move in opposite directions: PE10 and dividend yield show a very strong negative correlation ($\rho \approx -0.89$), while P/E ratio and dividend yield also correlate negatively ($\rho \approx -0.35$).
- P/E ratio is moderately correlated with volatility ($\rho \approx 0.40$), suggesting that high valuations often coincide with higher market uncertainty.
- Other relationships are weak. We can indicate that returns, volatility, and valuations capture largely complementary dimensions of market behavior.

Correlations highlight strong overlaps between returns themselves and between valuation and dividend yield, while other features remain relatively independent.

3.4 Hypothesis Testing

We focus on volatility as a primary measure of market risk. The hypothesis is formulated as:

- H_0: Volatility during crisis periods equals volatility during non-crisis periods.
- H_1: Volatility during crisis periods is greater than during non-crisis periods.

	Count	Mean	Std	Min	Median	Max
Crisis	64	0.389	0.172	0.088	0.341	0.686
Non-crisis	458	0.259	0.105	0.054	0.253	0.580

Table 3.6: Comparison of 12-month volatility between crisis and non-crisis periods

Using a one-sided Welch's t-test, we compare the two groups. The descriptive statistics show that average volatility during crises is approximately 0.39 with higher dispersion (std ≈ 0.17), compared to a mean of 0.26 with lower dispersion (std ≈ 0.10) in non-crisis periods. The test yields a t-statistic of 5.89 with a one-tailed p-value < 0.001, leading us to reject the null hypothesis.

Figure 3.3: Boxplot and histogram of 12-month volatility during crisis and non-crisis periods

Conclusion: Volatility is significantly higher during crises, confirming that financial stress is most clearly reflected in sharp increases in market risk.

Chapter 4

Visualization

To get better insights into S&P 500 Dataset, several visualization techniques are applied. These figures provide an intuitive view of market dynamics and reveal how key indicators behaved under both normal conditions and financial crises

4.1 Market Performance

Figure 4.1: S&P 500 Index with Highlighted Crisis Periods (1980–2023)

As shown in Figure 4.1, The S&P 500 shows strong long-term growth but sharp declines during crises.

- Black Monday in 1987 produced a sudden sharp decline,
- The dot-com crash (2000–2002) and the Global Financial Crisis (2008) resulted in prolonged downturns.
- The COVID-19 shock triggered an abrupt fall, followed by a rapid rebound.

These disruptions illustrate how quickly markets can collapse. Crises create extreme stress and challenge assumptions of growth and stability.

Figure 4.2: Monthly vs Real Return (2000–2009)

In Figure 4.2, both monthly and real returns display high volatility during the 2000–2009 period, which covers the dot-com crash (2000–2002) and the Global Financial Crisis (2008).

- Real returns often drop deeper than nominal returns, especially during crisis years.
- Sharp negative spikes in 2000–2002 and 2008 highlight how inflation-adjusted outcomes can be more severe.

Evaluating performance without accounting for inflation may underestimate the true impact of crises. Therefore, crises challenge assumptions of growth and stability, requiring careful consideration of real returns for investment decisions.

4.2 Valuation Metrics

Figure 4.3: Scatter Plot: PE10 vs Dividend Yield

From Figure 4.3, there is a clear negative relationship between the cyclically adjusted PE10 and dividend yield. We can conclude that when valuations are stretched, the income cushion from dividends diminishes. This directly supports the problem statement that examining PE10 and dividends under stress helps investors assess vulnerabilities and prepare for downturns.

Figure 4.4: PE10 over time

In Figure 4.4, the PE10 rises sharply during expansionary phases but then quickly contracts once crises unfold.

- Valuations peaked during the late 1990s dot-com bubble, reaching historically extreme levels (≈ 48) before collapsing as the dot-com bubble burst.
- It fell again during the Global Financial Crisis, and spiked once more around COVID-19 before adjusting downward.

Extreme valuations rarely persist and tend to reverse sharply under stress, supporting hat PE ratios can serve as early signals of market vulnerability.

4.3 Risk

Figure 4.5: 12-Month Volatility over time

Figure 4.5 shows the 12-month rolling volatility clearly spikes during financial crises.

- Volatility surged during the dot-com crash, reached extreme levels (close to 70%) in the Global Financial Crisis, and rose again during the COVID-19 shock.
- Between crises, volatility gradually declined as markets stabilized, highlighting its cyclical nature.

These recurring surges demonstrate how crises generate extreme uncertainty, disrupt stability and reinforcing that volatility is a key measure of market stress and risk exposure.

4.4 Dividends

Figure 4.6: Dividend Yield with Highlighted Crisis Periods (1980–2023)

As shown in Figure 4.6, the dividend yield shows a long-term declining trend but with noticeable spikes during crises.

- Yields were above 5% in the early 1980s but steadily declined to near 1% around the dot-com crash, reflecting elevated valuations and reduced income returns.
- During the Global Financial Crisis, yields briefly rebounded above 3% as stock prices fell sharply.
- In the COVID-19 shock, yields rose only modestly, staying below earlier peaks.

While dividends provide some cushion during downturns, their effectiveness has weakened over time as yields have trended lower. This directly supports the problem statement that analyzing dividends under stress is critical for understanding market resilience and investor income security.

Conclusion

Through the analysis of the S&P 500 index from 1980 to 2023, this study shows how financial crises have shaped market performance, valuation, and risk. The results show that while the index demonstrates long-term growth, crisis periods are characterized by extreme volatility, sharp valuation corrections, and temporary spikes in dividend yields. These patterns confirm that financial crises create significant stress in equity markets, challenging assumptions of stability and long-term trends.

The findings provide not only a descriptive picture of historical market behavior but also practical implications. Investors can use insights from valuation ratios and volatility as early warning signals for future downturns, while policymakers may rely on these indicators to better understand systemic risks. Moreover, the derived features—such as real returns, dividend yield, and rolling volatility—create a strong base for quantitative modeling and forecasting of market resilience.

Overall, this project demonstrates the value of combining data engineering with statistical and visualization methods. By transforming historical financial data into structured insights, the study not only documents how the S&P 500 behaves under crises but also provides lessons that can support decision-making in risk management, portfolio strategy, and economic policy.

References